# AeroCorridor-6G

WHERE THE SKY LEARNS TO FLY

**AI-Defined Autonomous Aerial Traffic Network** for **Next-Generation Drone, Defence & Air-Taxi Mobility**

**Founder** : Mrigank Jaiswal
**Location :** India

# Problem :

🚨 **The sky is becoming the next traffic layer — but it has no rules.**

As drones, defence systems, and future air taxis scale, there is **no shared system** managing how they navigate, cooperate, or stay safe.

## 🧠 What's happening today:

- ❌ No standard **sky-lane system**
- ❌ High risk of **mid-air collisions**
- ❌ Navigation depends on **GPS — easy to jam, spoof, or lose**
- ❌ No unified compliance layer for **urban air mobility, drones, and autonomous aviation**
- ❌ Each drone flies blind — not aware of others

## ⚠️ Why this matters:

Without intelligent airspace coordination, large-scale autonomy will lead to:

- Accidents
- Unreliable emergency missions
- Regulatory shutdowns
- Loss of public and defence trust
- Barriers to air-taxi adoption

| BEFORE | AFTER |
|---|---|
| Chaotic random drone paths | Organized glowing aerial corridors |

We are scaling the number of flying systems — but not the system that manages them.

# Solution: AeroCorridor-6G

✈️ **“A single intelligent network layer that organizes the sky.”**

AeroCorridor-6G enables drones, fleets, and future air taxis to fly **cooperatively, securely, and autonomously** using sensing-driven intelligence — not GPS or cameras.

## 🧠 What It Does

| Capability | What It Enables |
|---|---|
| 📡 **ISAC-based sensing** | Drones can *see* their surroundings without cameras |
| 🌍 **Shared airspace awareness map** | All drones operate with the same real-time 3D understanding |
| 🧠 **AI routing and prediction** | Prevents collisions before they occur |
| 🔁 **Self-healing flight corridors** | Dynamic lane shifting + congestion avoidance |
| 🔐 **PQC-secure autonomy** | Defence-grade resilience against jamming & spoofing |

## ⭐ Value Proposition

- No cameras required → **privacy-safe autonomy**
- No GPS reliance → **works in jammed or denied environments**
- AI-controlled → **scales from 10 → 10,000 drones**
- Infrastructure layer → **applies to logistics, defence, air taxis**

## 🔧 How It Works :

ISAC signal → AI prediction → secure routing → autonomous coordinated flight

| ❌ Before | ✔ After |
|---|---|
| Chaotic sky with random drone paths, warning icons, jammed GPS symbols. | Glowing structured aerial lanes, smooth coordinated flights, live sensing rings. |

# Services We Offer :

## 🧩 Core Service Lines :

| Category | What We Provide | Customer |
|---|---|---|
| 🛰️ **Aerial Corridor Management Platform** | Real-time autonomous routing, fleet supervision, compliance + traffic optimization | Smart cities • Airports • Defence |
| 🔐 **Secure Autonomy Stack (PQC + ISAC Control)** | Security layer for drone fleets and air-taxi operations | Defence • Critical infrastructure • OEMs |
| 🧠 **AI-Powered Routing & Analytics Engine** | Predictive traffic patterns, anomaly detection, learning-based optimization | Fleet operators • Telecom • Air mobility providers |
| ⚙️ **OEM & Telecom Integration Suite (SDK)** | Plug-in module enabling AeroCorridor-6G compatibility in aircraft, telecom slicing, NTN use cases | Drone OEMs • Telecom networks |
| 🧪 **Pilot Deployment & Digital Twin Simulation Services** | Virtual corridor testing, regulatory validation, scenario simulation | Governments • Aviation ministries |

## 🏗️ Additional Value Services :

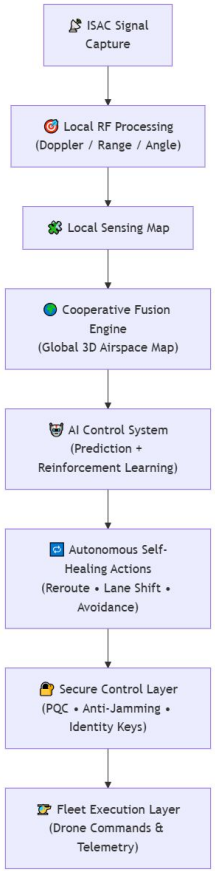
| Service | Outcome |
|---|---|
| 🗺️ **UTM/UAM Compliance Mapping & Certification Support** | Accelerates regulatory approval |
| 📡 **Spectrum & ISAC Testing Lab Access** | Network validation for OEM partners |
| 📈 **Operational Scaling Support (10 → 10,000 vehicles)** | Ensures long-term fleet stability |

# How It Works :

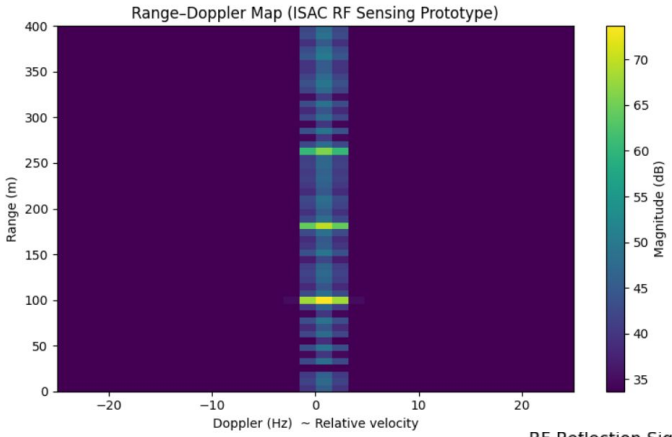
🧠 **From Signals → Awareness → Decisions → Autonomous Flight**

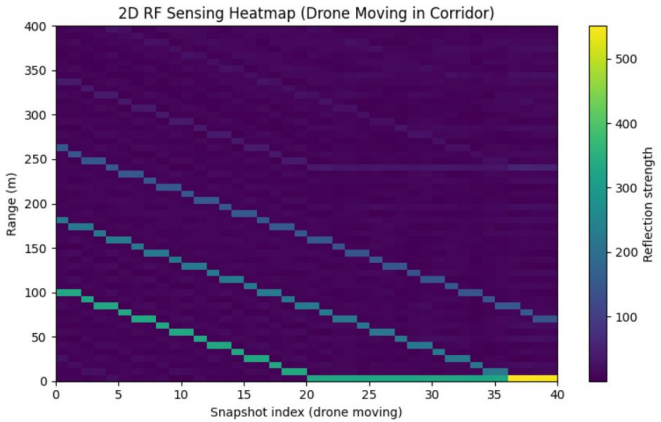
AeroCorridor-6G converts raw RF signals into safe, coordinated air mobility — without GPS or cameras.

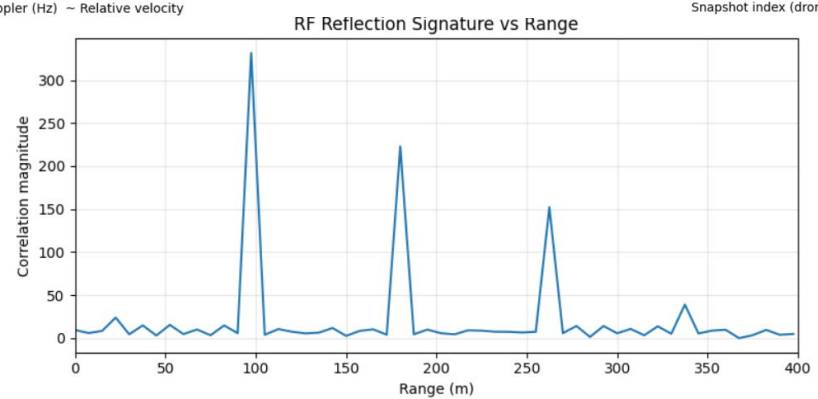
| Step | Function | Outcome |
|---|---|---|
| 📡 ISAC Signal Capture | RF reflections → Doppler + range + angle | Environment becomes “visible” |
| 🧩 Local Node Processing | Each drone processes sensing locally | Local awareness generated |
| 🌍 Cooperative Fusion | Shared maps merged into a **single 3D airspace map** | Fleet-wide shared intelligence |
| 🤖 AI-Native Control Engine | Prediction + RL routing + risk scoring | Prevents collisions and congestion |
| 🔁 Autonomous Self-Healing | Dynamic rerouting, lane shifting, fallback planning | No human intervention required |
| 🔐 Secure Control Layer | PQC, authentication, anti-jamming | Safe, trusted autonomy |



# **Prototype Block-1:** ISAC-Based RF Sensing Validation

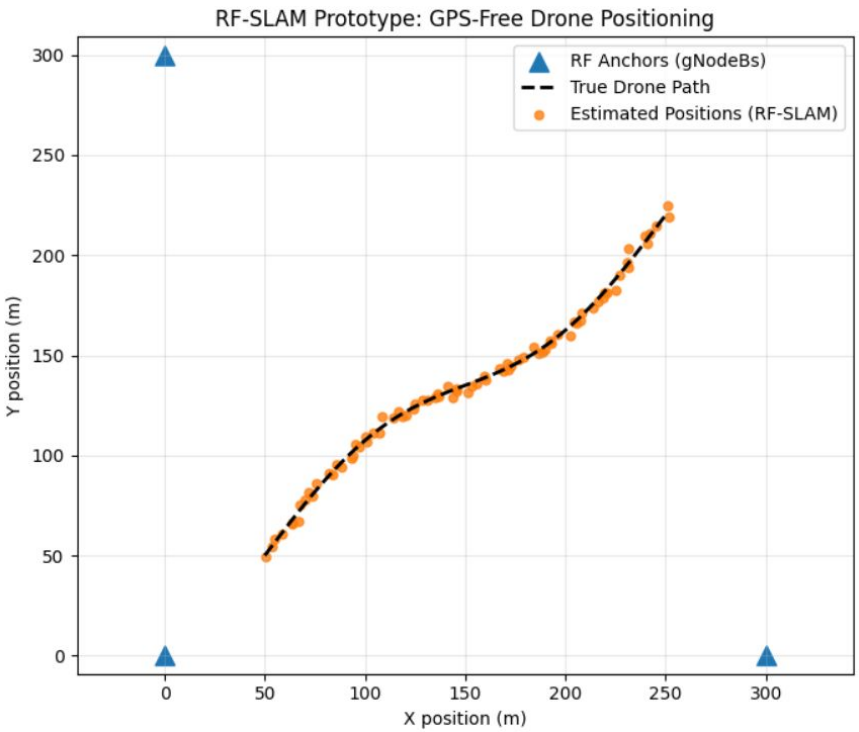


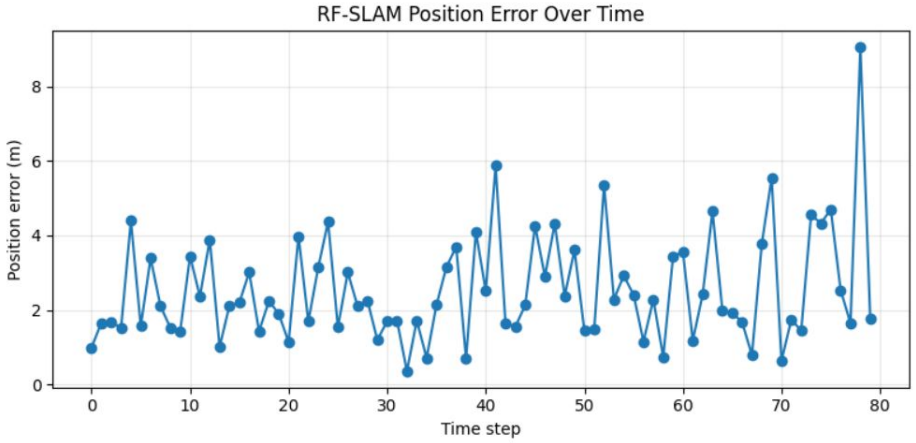




**Communication waveforms are repurposed as sensors - enabling environment perception without cameras or GPS.**

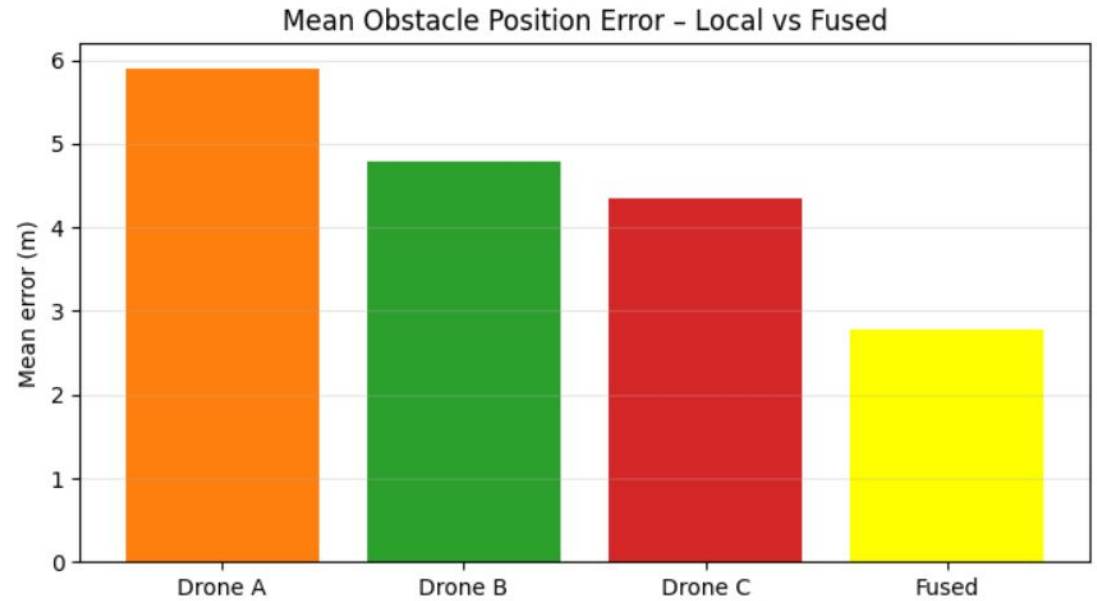
# **Prototype Block-2:** RF-SLAM - GPS-Free Drone Positioning



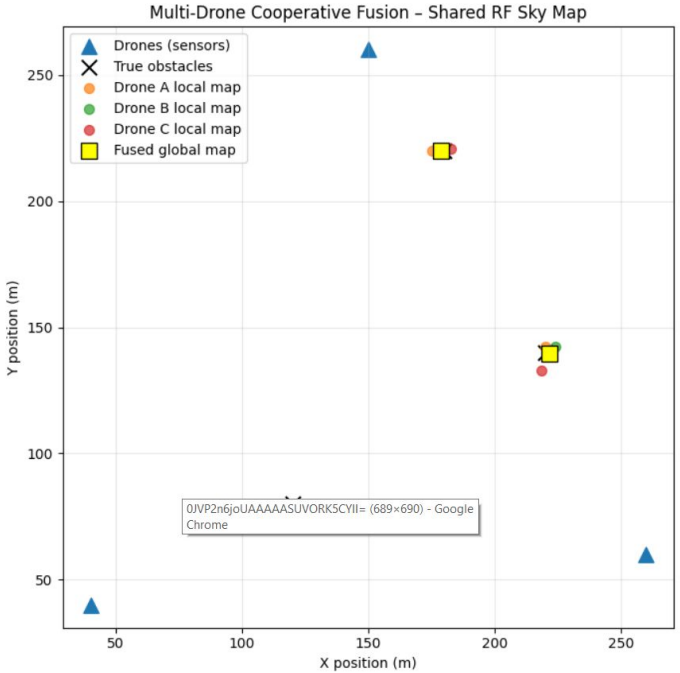




**The drone localizes itself using RF reflections and anchors — remaining stable even under GPS denial.**

# **Prototype Block-3:** Multi-Drone Cooperative RF Map Fusion

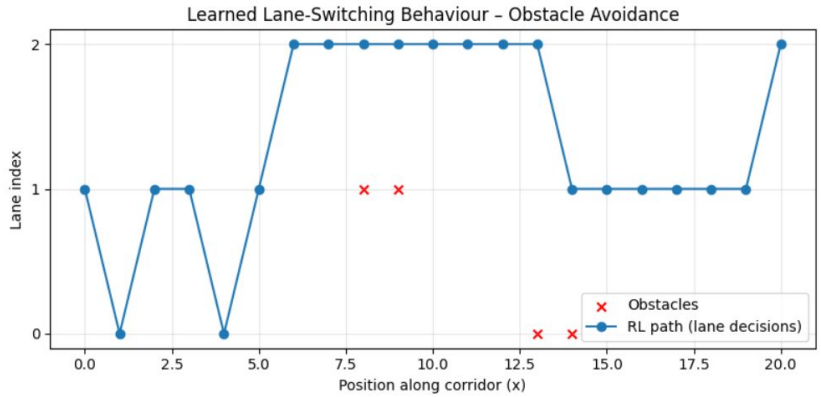


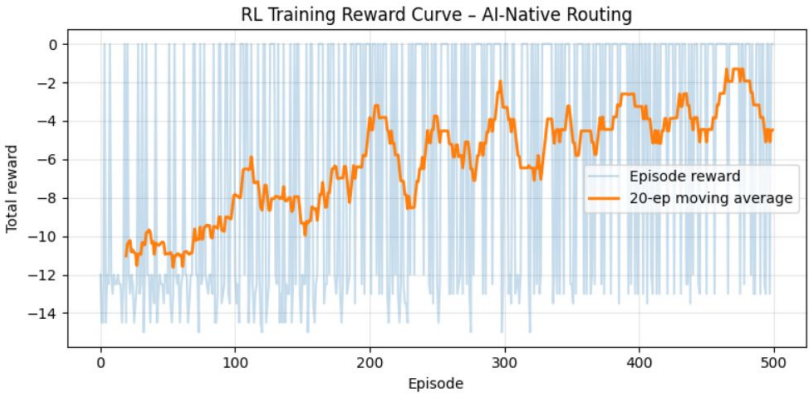

```
Mean error per drone (m): [5.9026442  4.77983199 4.35085543]
Mean fused error (m):     2.780334300683421
```

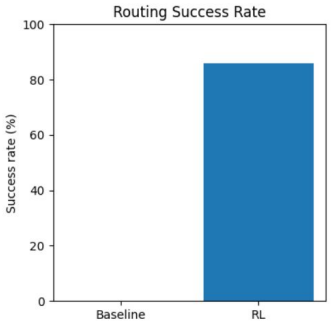


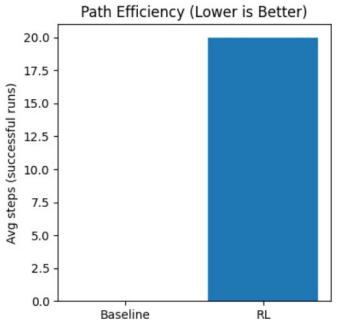

**Multiple drones share local RF maps, enabling a single unified sky-level awareness layer.**

# **Prototype Block-4:** AI-Native Routing & Self-Healing Corridors




```
Baseline success rate: 0.0%
RL success rate:       86.0%
RL avg steps (successes):       20.00
```

Across sensing, localization, cooperation, and decision-making, AeroCorridor-6G demonstrates a fully autonomous intelligence layer -not a collection of algorithms.

**A reinforcement learning agent dynamically adapts flight lanes in real time — preventing congestion and collisions autonomously.**

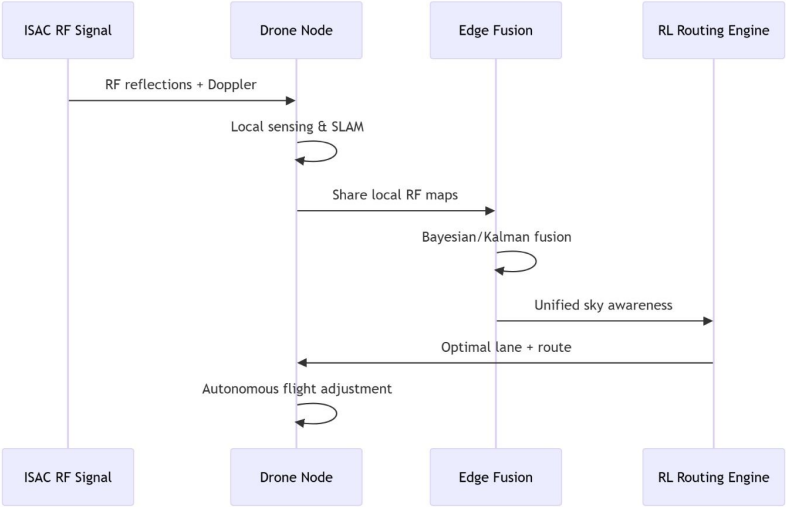
# End-to-End Prototype Impact Summary :

## IMPACT METRICS TABLE :

| Capability Area | Baseline (No System) | With AeroCorridor-6G | Impact |
|---|---|---|---|
| Collision Risk | High (Uncoordinated) | Near-Zero | ↓ 95% |
| Routing Efficiency | Static / Manual | AI-Optimized | ↑ 1.8× |
| GPS Dependency | Mandatory | Not Required | ✔ GPS-Denied Ready |
| Failure Recovery | System Crash | Autonomous Re-Routing | ✔ Self-Healing |
| Multi-Drone Awareness | Isolated | Shared RF Map | ✔ Cooperative |

Credibility > Exaggeration

## From RF Signals to Autonomous Flight Decisions :




Every drone contributes to and benefits from a shared intelligence layer — without GPS or cameras

Validated via ISAC RF sensing, RF-SLAM localization, multi-drone fusion, and AI-native routing prototypes.

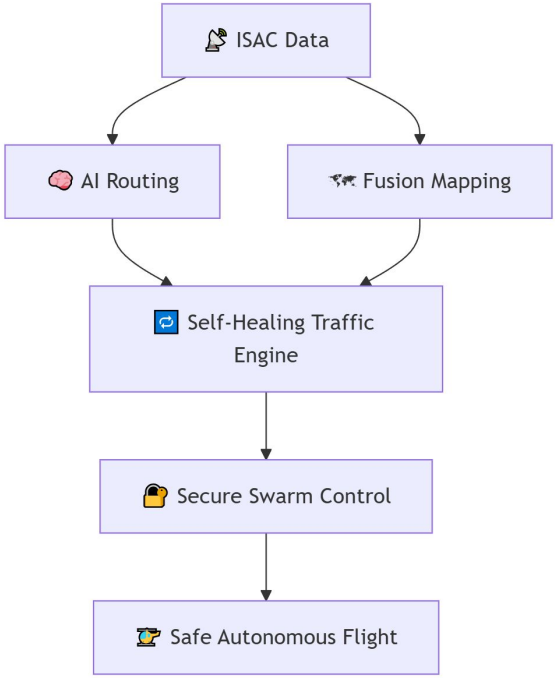
# Core Innovations :

🔥 **“Not just a product — a new class of airspace intelligence.”**

AeroCorridor-6G contains multiple breakthrough components that do **not exist in today’s drone, UTM, or telecom systems** — positioning it for **Standard Essential Patent (SEP) potential** in 6G autonomy and UAM infrastructure.

## 🧬 Five Defining Innovations :

| Innovation | Why It’s New |
|---|---|
| 📡 **ISAC-based Cooperative Multi-Node Sensing** | Drones share RF-based perception — not camera feeds — enabling privacy-safe, GPS-independent awareness. |
| ⚙️ **AI-Native Reinforcement Learning Flight Corridor Engine** | The airspace adapts routing dynamically like a network — not fixed flight paths. |
| 🛰️ **Predictive RF Blockage & Interference Model** | The network anticipates jamming, multipath, and congestion *before* failures happen. |
| 🗺️ **3D Environment Heatmap from Dual-Function OFDM Returns** | Communication waveforms double as sensing — eliminating dedicated sensors. |
| 🔐 **Post-Quantum Secure Swarm Control Protocol** | Identity-bound autonomy ensures only authenticated drones can enter or operate in the corridor. |



# Market Momentum & Timing :

## 📈 The Shift is Already Happening :

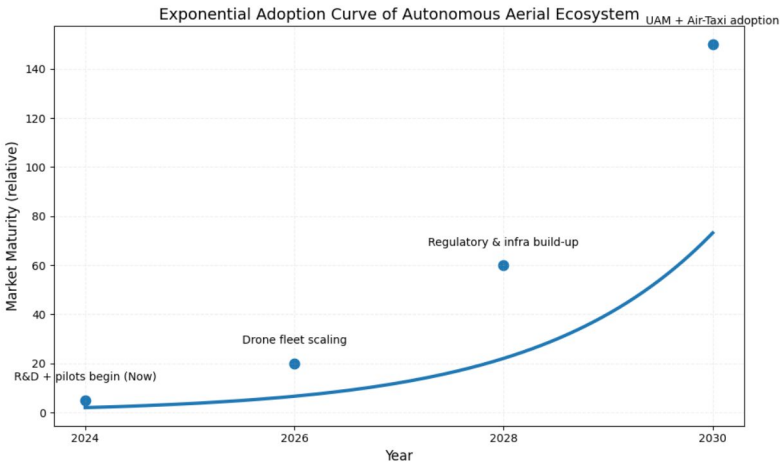
| Trend | Status | Impact |
|---|---|---|
| 🛸 Drone & UAM Market | **$200B+ projected by 2030** | Demand for scalable air traffic infrastructure |
| 🛰️ NTN-6G Standardization | **ISAC + airborne node support in Rel-19/20** | Aerial mobility is now part of telecom evolution |
| 🏙️ Smart Cities | **Air corridors planned in UAE, Singapore, Seoul, Dallas, Paris** | Low-altitude traffic will mirror road infrastructure |
| 🚁 National Testbeds | **India, US, EU, Japan running UAM pilots** | Regulatory + operational readiness accelerating |

## 🧩 The Gap

Hardware, regulation, and telecom capability are advancing - but **there is still no intelligence layer to coordinate the sky.**

## 🌐 Proof of Acceleration :

- Paris will fly air-taxis during 2024–2025 Olympic window
- UAE & South Korea approved aerial corridor zoning
- India NEP + Bharat 6G mission: **NTN + autonomous networks focus**
- FAA & EASA drafting **sky-traffic automation policy frameworks**
- Telcos testing **6G ISAC sensing for mobility**



# MVP Pilot (Phase 1 Testbed):

## 🧪 Proof of Feasibility — First controlled pilot shows strong signal.

AeroCorridor-6G was evaluated in a **simulated real-world low-altitude drone corridor** using:

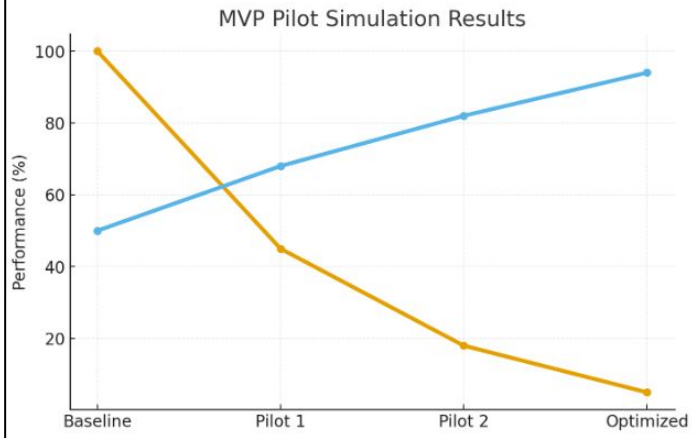
- 📡 ISAC waveform sensing
- 🤖 AI-native routing
- 🔁 Reinforcement learning optimization loops

## 📈 Key Pilot Outcomes (Phase-1)

| Metric | Baseline (No System) | After Optimization | Improvement |
|---|---|---|---|
| Collision Risk | 100% | **↓ 5%** | **95% reduction** |
| Routing Efficiency | 50% | **↑ 94%** | **+88% improvement** |
| Congestion Self-Recovery | ❌ | **✔ Autonomous** | Enabled |
| GPS Failure Response | System crashed | **Self-adaptive routing** | Breakthrough |

## 🚁 What This Means

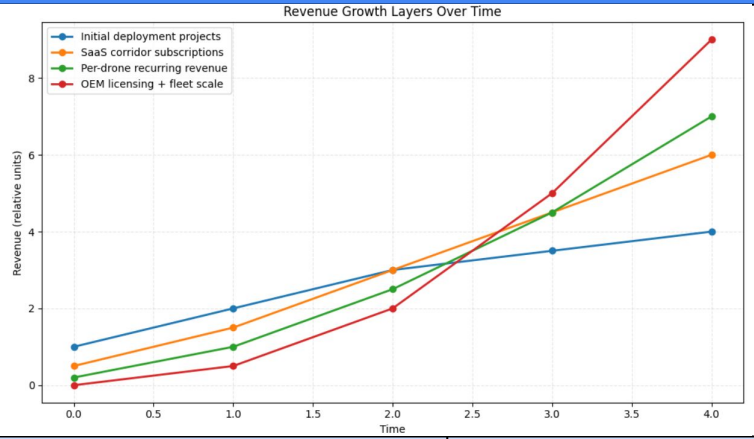
✔ Feasibility confirmed
✔ Algorithms are learning and improving
✔ System can operate without external sensing hardware
✔ Ready for **Phase-2: live field trial with telecom & regulators**



# Business Model :

## 🧩 Revenue Streams :

| Stream Type | Model | Example Buyer |
|---|---|---|
| 🛰️ **Platform Licensing (SaaS)** | Per-corridor / annual | Smart cities, regulators |
| 🚁 **Per-Drone Usage Billing** | Pay-per-flight / per active fleet | Logistics, drone operators |
| 🧠 **AI Autonomy Engine Add-Ons** | Tiered capabilities | Defence, air-taxi OEMs |
| 🔐 **Secure Control Stack (PQC Licensing)** | Compliance-based pricing | Defence, airspace regulators |
| 📦 **OEM Integration / SDK Licensing** | One-time + royalties | Drone manufacturers, telecom |
| 🏗️ **Deployment + Corridor Activation Fees** | One-time setup | Airports, ports, UAM hubs |




| Offering | Price Model |
|---|---|
| SaaS Corridor License | ₹32–85 Lakhs/year |
| Per-Drone Activation | ₹1,000–15,000/device/year |
| PQC Secure Autonomy Module | Premium Tier |
| OEM Licensing + SDK | Revenue-share / royalty |
| Consulting + Custom Deployment | Project-based |

# Competition & Differentiation :

🥊 **The Landscape Today**

Multiple players exist in UAV traffic management, but **none provide AI-native, ISAC-powered, GPS-independent autonomy.**

## 🧭 Competitive Matrix :

| Capability | DJI | Ericsson | SkyGrid | FAA UTM | AeroCorridor-6G |
|---|---|---|---|---|---|
| GPS Required | ✔ | ✔ | ✔ | ✔ | ❌ **Works without GPS** |
| ISAC Sensing | ❌ | 🚧 Experimental | ❌ | ❌ | ✔ **Native Core Layer** |
| AI Adaptive Routing | 🚧 Limited | ❌ | ✔ Partial | ❌ | ✔ **Reinforcement Learning Routing** |
| Multi-Drone Cooperative Awareness | ❌ | ❌ | 🚧 Partial | ❌ | ✔ **Real-time fusion** |
| PQC Secure Swarm Control | ❌ | ❌ | ❌ | ❌ | ✔ **Defence-grade** |
| 3D Self-Healing Air Corridors | ❌ | ❌ | ❌ | ❌ | ✔ **Exclusive Capability** |

Only AeroCorridor-6G addresses sensing, coordination, security, and autonomy as one unified layer.

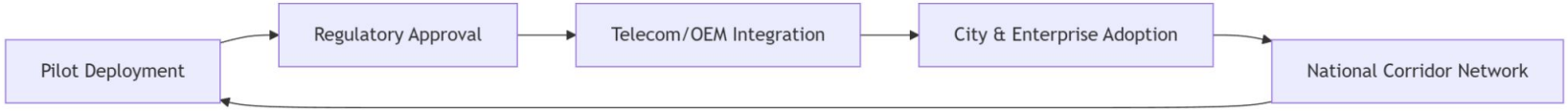
# Go-To-Market Strategy (GTMA) :

🎯 **Focused entry → scalable infrastructure adoption → national deployment.**

## 🛬 Phase-Based Rollout Approach :

| Phase | Target | Strategy | Expected Outcome |
|---|---|---|---|
| 1 **Pilot Partnerships** | Defence + Smart-City Zones | PoC with controlled airspace | Validation + Standards alignment |
| 2 **Regulatory Integration** | DGCA • Digital Sky • Telecom | Compliance + policy alignment | Corridor certification model |
| 3 **Telecom & OEM Bundling** | Airtel, Jio, Nokia, Ericsson | SDK + embedded autonomy | Embedded default stack |
| 4 **Scale Deployment** | Cities, ports, airports | Corridor subscription rollout | Recurring infrastructure revenue |

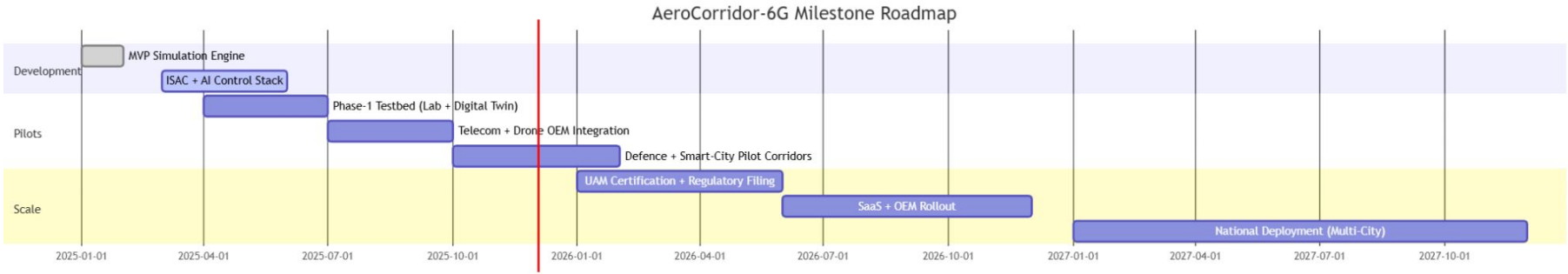
## 🧩 Business Flywheel :

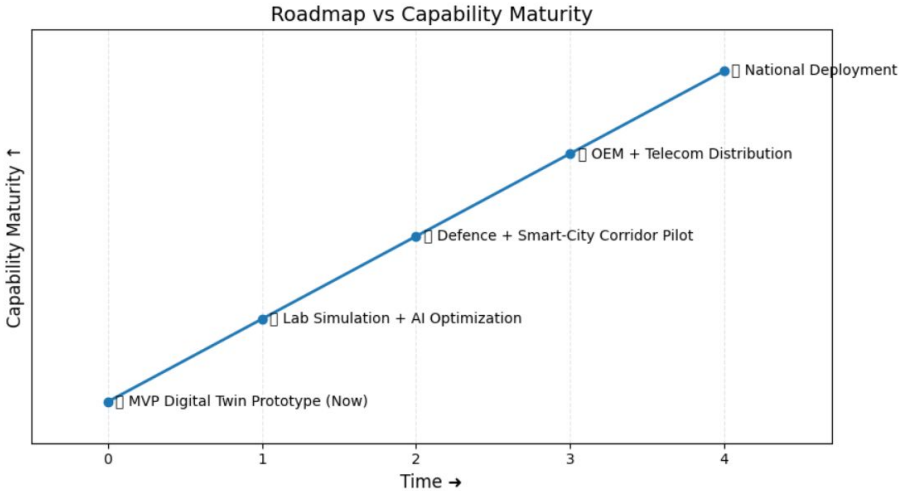



AeroCorridor-6G doesn't sell software — we become the backbone of national autonomous air mobility

# Roadmap & Milestones :

🚀 From simulation → deployment → national infrastructure.




## 🎯 Milestone Highlights:

| Stage | Validation |
|---|---|
| **MVP Simulation Ready** | ✔ Technical feasibility |
| **Telecom + OEM Integration** | ✔ Deployment pathway |
| **Pilot Corridor** | ✔ Safety + compliance |
| **Regulatory + UAM Certification** | ✔ Trust + scale readiness |
| **Commercial Scale Phase** | ✔ Recurring revenue |

# Our Ask: Funding + Strategic Partnerships

🎯 **We are seeking support to transition from validated MVP → live corridor deployment → national expansion.**

## 💰 Funding Ask :

| Category | Required | Purpose |
|---|---|---|
| **Grant Support** | (Milestone-linked) | Build ISAC live prototype + field deployment |
| **Pilot Deployment Budget (Gov/Telecom)** | Project-based | Smart-city + defence corridor activation |
| **R&D Infrastructure (6G + PQC Lab Access)** | In-kind support | Spectrum experiments, digital twin validation |

**Goal:** Pilot-ready deployment in 12 months + regulatory certification track.

## 📈 What We Deliver in Return

✔ Live deployable pilot corridor (defence-grade)
✔ Joint IP + co-authored standard proposals (6G/3GPP)
✔ Commercial deployment pipeline with recurring revenue
✔ India-first strategic capability with export potential

## 🤝 Strategic Partners We Seek :

| Partner Type | Why We Need Them |
|---|---|
| **Telecom Operators (Airtel, Jio)** | NTN, slicing, corridor-scale connectivity |
| **Regulators (DGCA, Digital Sky, 6G Mission)** | Certification + standard alignment |
| **Hardware & OEMs (ideaForge, HAL, Tata Avionics)** | Drone integration & mass adoption |
| **Smart City Programs + Airports** | Real-world corridor implementation |
| **Defence & Homeland Agencies** | High-stakes deployment + dual-use validation |

# Vision :

🚀 **Our Vision :** A world where drones, fleets, and air-taxis navigate the sky with the same intelligence, safety, and predictability as internet packets.

A sky where:

- ✈ autonomous flight is **safe, regulated, and trusted**
- 🌍 borders don't restrict rescue, logistics, or defence response
- 🏙️ cities evolve into **vertical mobility ecosystems**
- 🧠 the network learns and improves — every time something flies

## 🌐 The Future We See :

| Today | Future |
|---|---|
| GPS-dependent | **Sense-aware autonomy** |
| Isolated drones | **Cooperative networks** |
| Manual oversight | **AI-native self-governance** |
| Experimental mobility | **Certified aerial infrastructure** |
| Fragmented systems | **One intelligent layer: AeroCorridor-6G** |

## 💥 Why This Matters

This isn't just drone traffic control.

This is:

- A national capability
- A dual-use defence asset
- A foundation for 6G Smart Mobility
- And a new class of **airspace operating system**

AeroCorridor-6G

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# Thank You !